

Localizations of Regular Local Rings are Regular

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Directed Reading Program

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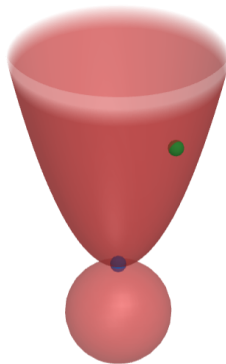
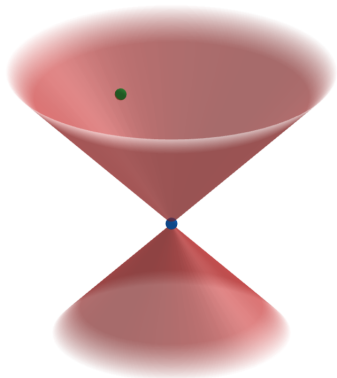
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Pictures



Algebraic surfaces defined by $x^2 + y^2 - z^2 = 0$ and $(x^2 + y^2 + 1 - z)(x^2 + y^2 + z^2 - 1) = 0$.

Introduction

Throughout this talk, we will assume that all rings are commutative and Noetherian, and that all modules are finitely generated.

Definition

A ring is **local** if it has a unique maximal ideal. When R is local with maximal ideal $\mathfrak{m}$, we may call the pair $(R, \mathfrak{m})$ a local ring.

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Definition

The **localization** R_P of a ring R at a prime P is defined to be the ring with elements as fractions $\frac{r}{u}$ where $r \in R$ and $u \in R \setminus P$ under the equivalence $\frac{r_1}{u_1} = \frac{r_2}{u_2}$ if there exists $v \in R \setminus P$ such that $v(r_1 u_2 - r_2 u_1) = 0$. (R_P, P_P) is a local ring.

Introduction

Definition

Given a prime ideal P of a ring R , we define the **codimension** of P , $\text{codim } P$, to be the supremum of all n such that there exists prime ideals $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n = P$.

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Example

In the ring $\mathbb{C}[x, y, z]$, the elements of a prime ideal P are polynomials. We can consider the 0-set $Z(P)$ of points in $\mathbb{C}^3$, that is, the set of points (x_0, y_0, z_0) such that $f(x_0, y_0, z_0) = 0$ for all $f \in P$. The codimension of P is equal to the dimension of the entire space $\mathbb{C}^3$ minus the dimension of $Z(P)$ (for the appropriate definition of “dimension”).

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Definition

A local ring $(R, \mathfrak{m})$ is **regular** if the minimal number of generators of $\mathfrak{m}$ is equal to $\text{codim } \mathfrak{m}$.

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A localization of a regular local ring is regular.

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Example

$\mathbb{R}[x, y]_{(x, y)}$ is regular local, with maximal ideal $(x, y)_{(x, y)}$ having codimension 2 and generated by $x/1$ and $y/1$. Localizing at the prime ideal $(x)_{(x, y)}$ gives the ring $\mathbb{R}[x, y]_{(x)}$, which has maximal ideal $(x, y)_{(x)}$ of codimension 1 generated by the one element $x/1$.

Regular Sequences and Depth

Definition

Let R be a ring, and let M be an R -module. A **regular sequence** on M , or an M -sequence, is a sequence of elements $x_1, x_2, \dots, x_n \in R$ such that $M/(x_1, \dots, x_n)M \neq 0$ and for $i = 1, \dots, n$, x_i is a nonzerodivisor on $M/(x_1, \dots, x_{i-1})M$. A **regular sequence** shall by default be on R when viewed as an R -module.

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Example

For any ring R , the sequence x, y, z is a regular sequence in the ring $R[x, y, z]$.

Regular Sequences and Depth

Definition

Over a ring R , given an ideal I and module M , we define the **depth of I on M** , $\text{depth}(I, M)$ to be the maximum possible length of an M -sequence, provided $IM \neq M$ (if $IM = M$, we shall define $\text{depth}(I, M) = \infty$). We define the **depth of I** to be the depth of I on R when viewed as a module over itself.

It will turn out that all M -sequences that cannot be extended must have length $\text{depth}(I, M)$.

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Example

In the ring $\mathbb{C}[t^2, t^3]$, the depth of (t^2, t^3) , which is a maximal ideal, is 1; examples of maximal regular sequences of this length are t^2 and t^3 .

Regular Sequences and Depth

Proposition

Given a ring R with prime ideal P , the depth of P is at most the codimension of P (and is thus well-defined as it is finite).

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Example

Recall the example of the maximal ideal (t^2, t^3) of the ring $\mathbb{C}[t^2, t^3]$ has regular sequences of length 1. This shows the depth is at least 1. We claimed that the depth was in fact equal to 1, and this can be seen from the fact that the ideal (t^2, t^3) has codimension 1.

Regular Sequences and Depth

Proposition

If R is a local ring with maximal ideal $\mathfrak{m}$, then R is regular if and only if $\mathfrak{m}$ is generated by a regular sequence (on R).

Furthermore, if R is regular, then any sequence of elements generating $\mathfrak{m}$ is regular if and only if it is of length $\text{codim } R$.

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In the regular local ring $\mathbb{R}[x, y, z]_{(x, y, z)}$, the maximal ideal is generated by $x/1, y/1, z/1$, which is indeed a regular sequence.

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Example

The ring $(\mathbb{R}[x, y]/(xy))_{(x, y)}$ is not regular as the codimension of the maximal ideal is 1 but the depth of the maximal ideal is 0, and it follows from the proposition that in any regular local ring the codimension must equal the depth for the maximal ideal.

Koszul Complexes

Definition

A **cochain complex** is a sequence of modules $\dots, M_{-1}, M_0, M_1, \dots$ with a set of homomorphisms $\varphi_n : M_n \rightarrow M_{n+1}$ s.t. $\varphi_{n+1} \circ \varphi_n = 0$. In other words, $\text{im}\varphi_n \subseteq \ker\varphi_{n+1}$.

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Definition

The **cohomology** H_n with respect to a cochain complex $\dots \rightarrow M_{n-1} \xrightarrow{\varphi_{n-1}} M_n \xrightarrow{\varphi_n} M_{n+1} \xrightarrow{\varphi_{n+1}} \dots$ is the quotient group $H_n = \ker\varphi_{n+1} / \text{im}\varphi_n$.

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Definition

An **exact sequence** is a cochain complex $\dots \rightarrow M_{n-1} \xrightarrow{\varphi_{n-1}} M_n \xrightarrow{\varphi_n} M_{n+1} \xrightarrow{\varphi_{n+1}} \dots$ s.t. $\forall n, \ker\varphi_{n+1} = \text{im}\varphi_n$.

Koszul Complexes

Definition

Given a ring R , a R -module M , the **exterior algebra** $\wedge M$ is defined as the free algebra $R \oplus M \oplus (M \otimes M) \oplus \cdots$, modulo the relations $x \otimes y = -y \otimes x$ and $x \otimes x = 0$.

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Example

If $R^n = \langle e_1, e_2, \dots, e_n \rangle$, for $M = R^n$, the following holds:

$$\bigwedge^m R^n = \langle e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_m} \mid 1 \leq i_1 < \cdots < i_m \leq n \rangle \cong R^{\binom{n}{m}}$$

Koszul Complexes

Definition

Given a ring R , a R -module M , and an element $x \in M$, the **Koszul Complex** $K(x)$ is the chain complex

$0 \rightarrow R \rightarrow M \rightarrow \wedge^2 M \rightarrow \cdots \rightarrow \wedge^i M \xrightarrow{d_x} \wedge^{i+1} M \rightarrow \cdots$, where d_x is defined by sending $a \in \wedge M$ to $x \wedge a$.

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Example

If $R^n = \langle e_1, e_2, \dots, e_n \rangle$, for $M = R^n$, then x can be written as $(x_1, \dots, x_n)$. We may write $K(x)$ in this case as $K(x_1, \dots, x_n)$, and it becomes

$$0 \rightarrow R^{\binom{n}{0}} \rightarrow R^{\binom{n}{1}} \rightarrow R^{\binom{n}{2}} \rightarrow \dots \rightarrow R^{\binom{n}{n-1}} \rightarrow R^{\binom{n}{n}} \rightarrow 0$$

Observe that all of these modules are free.

Koszul Complexes

Theorem

Let M be a module over a ring R . If $H^j(M \otimes K(x_1, \dots, x_n)) = 0$ for $j < r$ while $H^r(M \otimes K(x_1, \dots, x_n)) \neq 0$, then every maximal M -sequence in $I = (x_1, \dots, x_n) \subset R$ has length r .

From this theorem, all maximal M -sequences have lengths equal to the depth of M .

Koszul Complexes

Corollary

Let $x_1, \dots, x_n$ be an M -sequence. Then $M \otimes K(x_1, \dots, x_n)$ is exact until the last step. $H^n(M \otimes K(x_1, \dots, x_n)) = M/(x_1, \dots, x_n)M$.

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Remark

This results in a free resolution of $M/(x_1, \dots, x_n)M$.

Regular Local Rings

Definition

A **projective resolution** of an R -module M is a complex of projective modules

$$\mathcal{F} : \cdots \rightarrow F_n \xrightarrow{\varphi_n} \cdots \rightarrow F_1 \xrightarrow{\varphi_1} F_0$$

such that the sequence is exact at each F_i and $\text{coker } \varphi_1 = M$.

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Definition

The **projective dimension** $\text{pd } M$ of M is the minimum length of a projective resolution of M .

Definition

The **global dimension** of R is the supremum of the projective dimensions of all R -modules.

Regular Local Rings

Definition

Over a local ring $(R, \mathfrak{m})$, a **minimal complex** is a complex

$$\mathcal{F} : \cdots \rightarrow F_n \xrightarrow{\varphi_n} F_{n-1} \rightarrow \cdots$$

such that the image of φ_n is contained in $\mathfrak{m}F_{n-1}$ for all n .

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Remark

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Remark

The number of generators of each F_i is minimal.

Lemma

Let R be a local ring with residue field $R/\mathfrak{m} = k$, and let $M \neq 0$ be an R -module. Then every minimal free resolution of M has length $\text{pd } M$, and the global dimension of R is $\text{pd } k$.

Regular Local Rings

Corollary

Let R be a regular local ring with dimension n . The global dimension of R is n .

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Proof Idea:

From the previous lemma, the global dimension of R is $\text{pd } k$, and the Koszul complex of the generators of the maximal ideal gives a minimal free resolution of k .

Regular Local Rings

Auslander-Buchsbaum Formula

Let R be a local ring with maximal ideal $\mathfrak{m}$, and let M be an R -module of finite projective dimension. Then

$$\text{pd } M = \text{depth}(\mathfrak{m}, R) - \text{depth}(\mathfrak{m}, M).$$

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Let R be a local ring with maximal ideal $\mathfrak{m}$, and let M be an R -module of finite projective dimension. Then

$$\text{pd } M = \text{depth}(\mathfrak{m}, R) - \text{depth}(\mathfrak{m}, M).$$

Proof

We proceed by induction on $\text{pd } M$. If $\text{pd } M = 0$, M is free and $\text{depth}(\mathfrak{m}, R) = \text{depth}(\mathfrak{m}, M)$.

Otherwise, let

$$\varphi : F \rightarrow M$$

be the first map in a minimal free resolution of M , and let $N = \ker \varphi$. By a previous lemma, $\text{pd } N = \text{pd } M - 1$. Hence through induction, we may apply the formula to N .

Regular Local Rings

Auslander-Buchsbaum Formula

$$\text{pd } M = \text{depth}(\mathfrak{m}, R) - \text{depth}(\mathfrak{m}, M).$$

Proof (cont'd)

Let $d = \text{depth}(N)$. We wish to show that $\text{depth}(M) = d - 1$. Let $x = (x_1, \dots, x_n)$ be a set of generators of $\mathfrak{m}$, so that we have the Koszul complex $K(x)$. We also have a short exact sequence

$$\mathcal{F} : 0 \rightarrow N \rightarrow F \rightarrow M \rightarrow 0.$$

We tensor $K(x)$ with $\mathcal{F}$ to obtain a short exact sequence of complexes, which then results in a long exact sequence in homology:

$$\cdots \rightarrow H^{i-1}(F \otimes K(x)) \rightarrow H^{i-1}(M \otimes K(x)) \rightarrow H^i(N \otimes K(x)) \rightarrow \cdots$$

Regular Local Rings

Auslander-Buchsbaum Formula

$$\text{pd } M = \text{depth}(\mathfrak{m}, R) - \text{depth}(\mathfrak{m}, M).$$

Proof (cont'd)

Since N has depth d , $H^d(N \otimes K(x)) \neq 0$, and since F also has depth at least d , for all $i < d - 1$, $H^i(M \otimes K(x)) = 0$. We wish to show that $H^{d-1}(M \otimes K(x)) \neq 0$. Since we have the sequence

$$\cdots \rightarrow H^{d-1}(M \otimes K(x)) \rightarrow H^d(N \otimes K(x)) \rightarrow H^d(F \otimes K(x)) \rightarrow \cdots$$

it suffices to show that the map

$$H^d(N \otimes K(x)) \rightarrow H^d(F \otimes K(x))$$

is the zero map.

Regular Local Rings

Auslander-Buchsbaum Formula

$$\text{pd } M = \text{depth}(\mathfrak{m}, R) - \text{depth}(\mathfrak{m}, M).$$

Proof (cont'd)

This map is induced by the inclusion $\iota : N \rightarrow F$. If $\text{pd } N > 0$, by the formula, $d < \text{depth } R$, so $H^d(F \otimes K(x)) = 0$. If $\text{pd } N = 0$, N is free, so $H^d(N \otimes K(x)) = N \otimes H^d(K(x))$ and $H^d(F \otimes K(x)) = F \otimes H^d(K(x))$. Then the map induced by ι is $\iota \otimes 1$. Since $\mathfrak{m}$ annihilates all Koszul homology groups, we see that $\iota \otimes 1$ is the zero map by representing ι as a matrix with entries in $\mathfrak{m}$. $\square$

Regular Local Rings

Lemma

Let R be a local ring with maximal ideal $\mathfrak{m}$, and let $k = R/\mathfrak{m}$. If $x = (x_1, \dots, x_n)$ is a minimal set of generators for $\mathfrak{m}$, $K(x) = K(x_1, \dots, x_n)$ is a subcomplex of the minimal free resolution of k .

Regular Local Rings

Theorem

A local ring has finite global dimension if and only if it is regular.

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Proof.

The "if" direction is done by a previous corollary. We now suppose that R has finite global dimension. Let $k = R/\mathfrak{m}$, and let $x = (x_1, \dots, x_n)$ be a minimal set of generators for $\mathfrak{m}$. By the principal ideal theorem, $\dim R \leq n$. From our previous result on depth and codimension, $\dim R \geq \text{depth}(R)$. By the Auslander-Buchsbaum formula, $\text{depth}(R) = \text{pd } k$, and by the previous lemma, $K(x)$ is contained in the minimal free resolution of k . Hence $\text{pd } k \geq n$. □

Regular Local Rings

Theorem

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Proof.

Let R_P be the localization of a regular local ring R at a prime ideal P . By the previous theorem, it suffices to show that R_P has finite global dimension. Using a previous lemma, its global dimension is equal to the projective dimension of its residue field R_P/P_P . Since R is regular, R/P has a finite projective resolution over R . We localize this resolution to obtain a finite free resolution of R_P/P_P over R_P . □

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References

David Eisenbud, *Commutative algebra with a view toward algebraic geometry*, Springer Science+Business Media, New York, 2004.